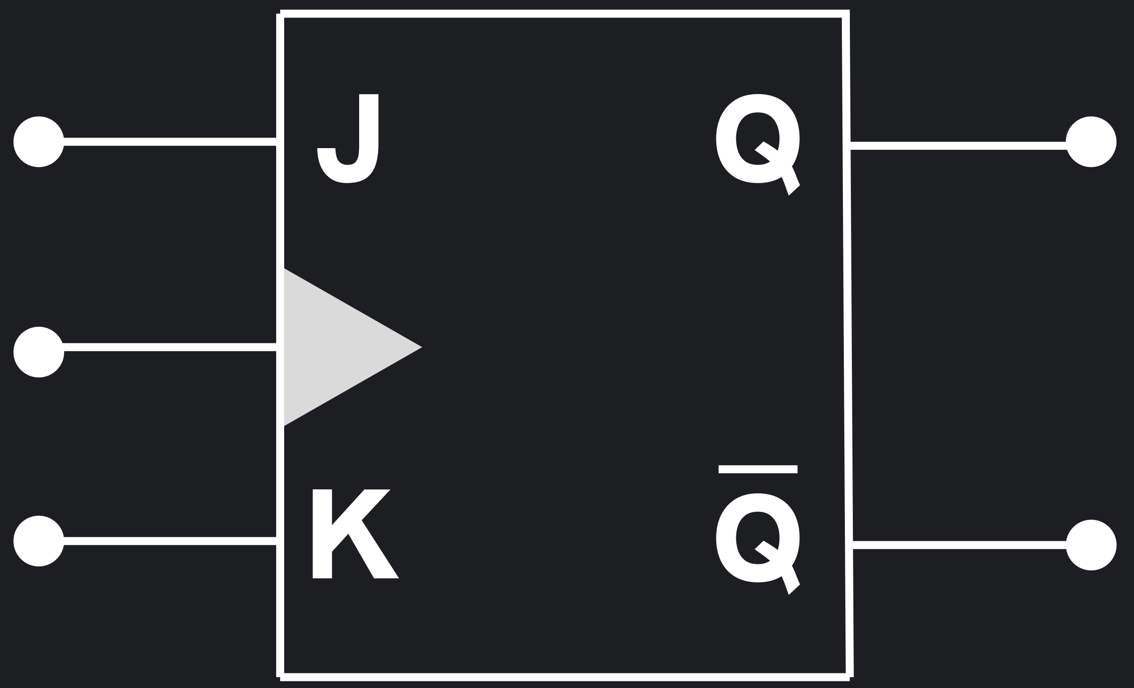
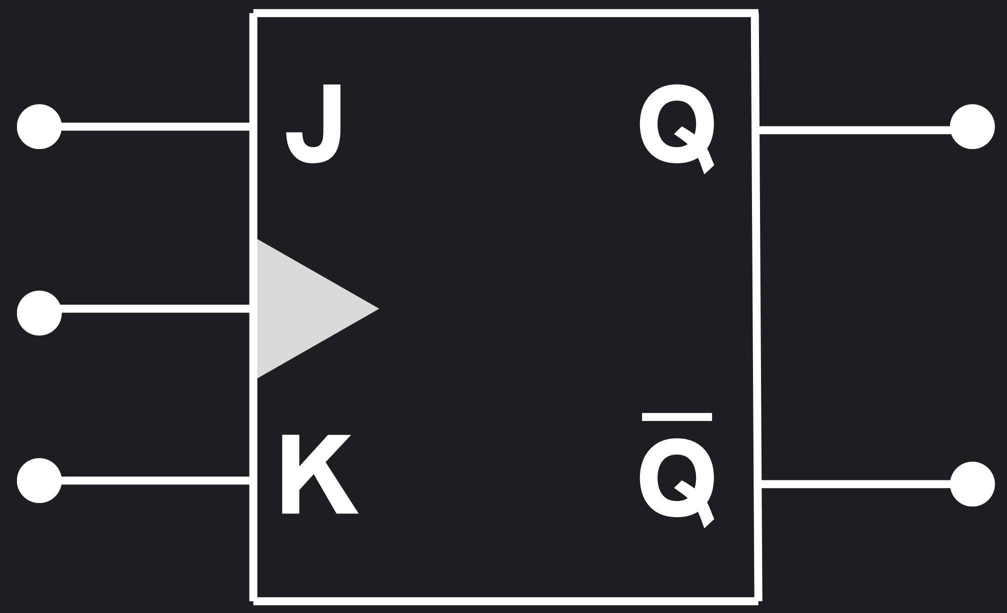
JK–FLIP FLOP



What is a JK-FLIP FLOP?

A JK Flip-Flop is a basic sequential logic gate that produces an output based on its inputs and its previous state. The output is in a Toggle state when both inputs (J and K) are HIGH (1), while it maintains its previous state when both inputs are LOW (0).

Truth Table



J	K	CK	Q	made of operation
x	x	$\uparrow$	Q_0	NO Change
0	0	$\downarrow$	Q_0	NO Change
0	1	$\downarrow$	0	Resets
1	0	$\downarrow$	1	Sets
1	1	$\downarrow$	Q_0	Change

Function of the JK-FLIP FLOP

A JK Flip-Flop performs a logical storage and Toggle operation on its inputs.The main function of a JK Flip-Flop is to serve as a basic memory element and produce an output signal according to this rule:

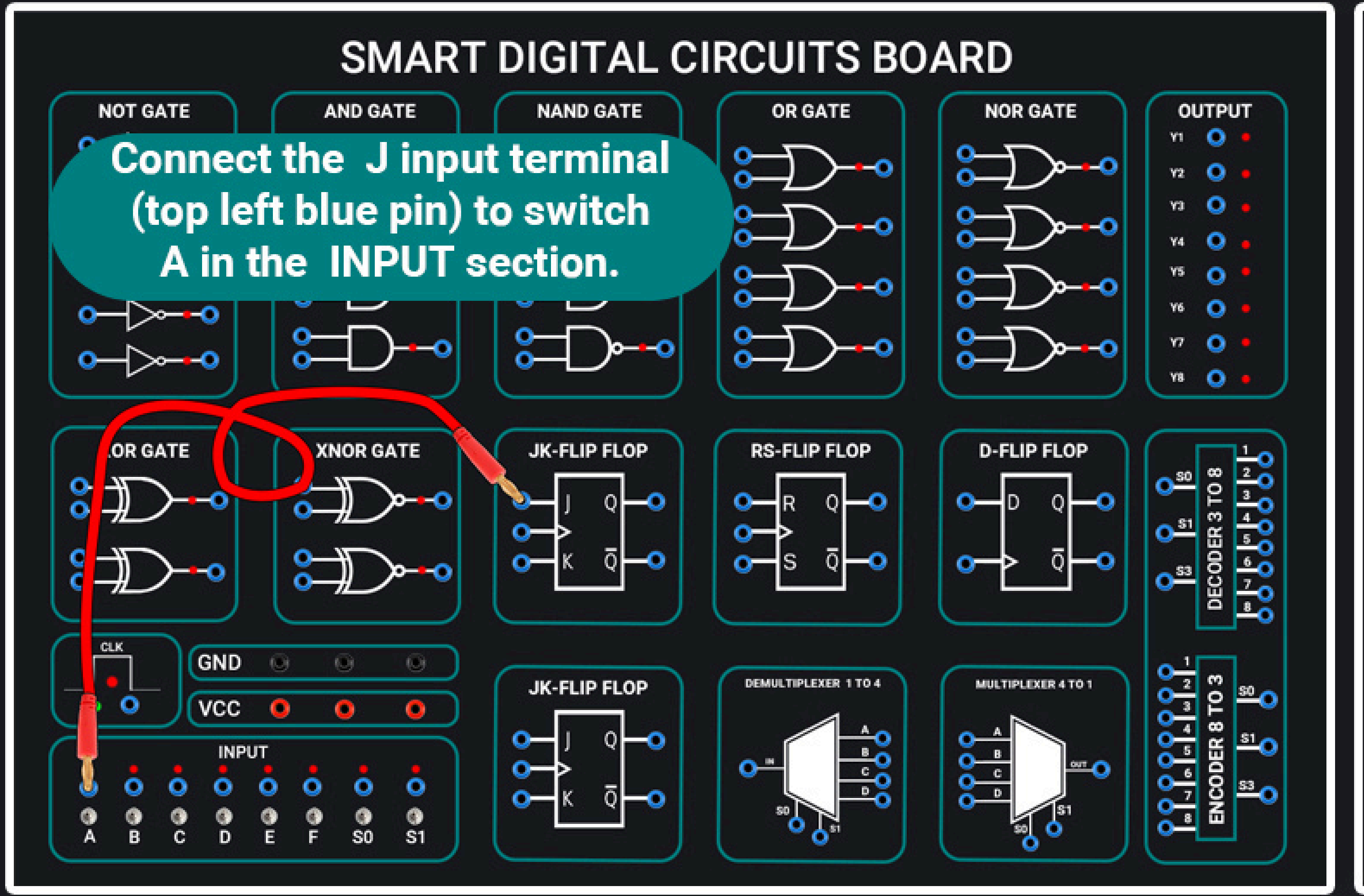
Output = Toggle when both inputs J and K = 1

Output = No Change when both inputs J and K = 0

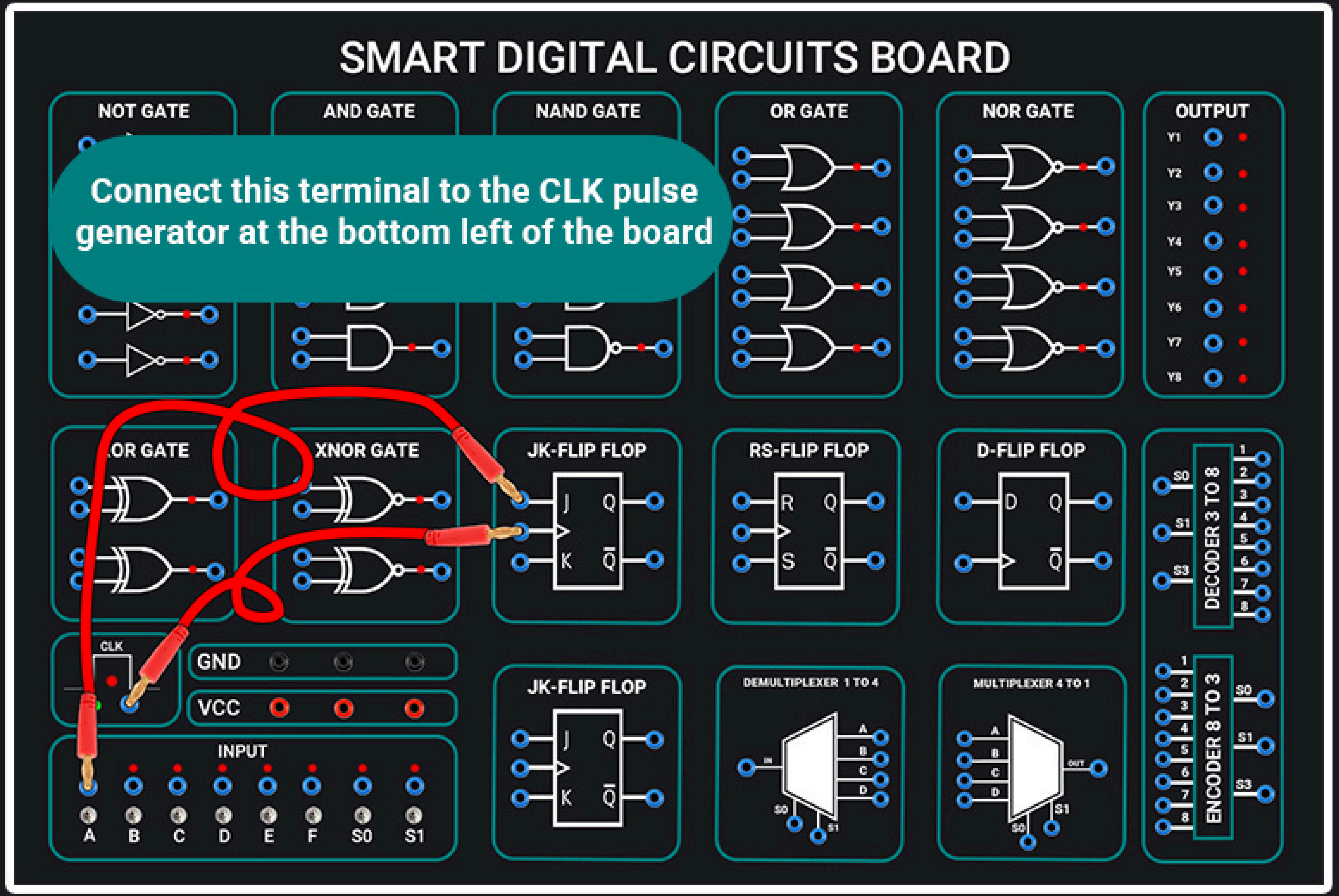
Output = 1 (Set) when J = 1 and K = 0

Output = 0 (Reset) when J = 0 and K = 1

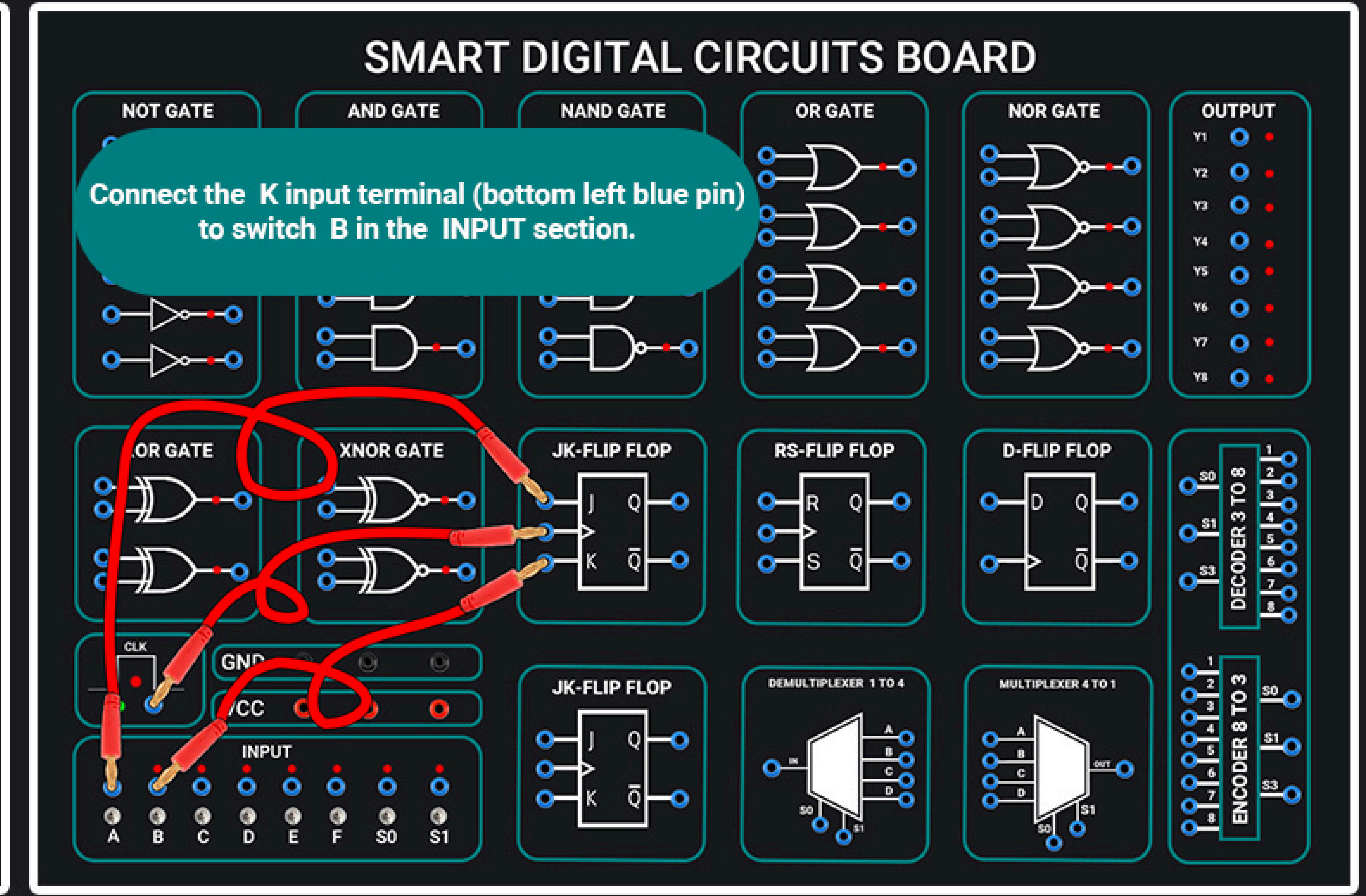
1


SMART DIGITAL CIRCUITS BOARD
Connect the J input terminal (top left blue pin) to switch A in the INPUT section.

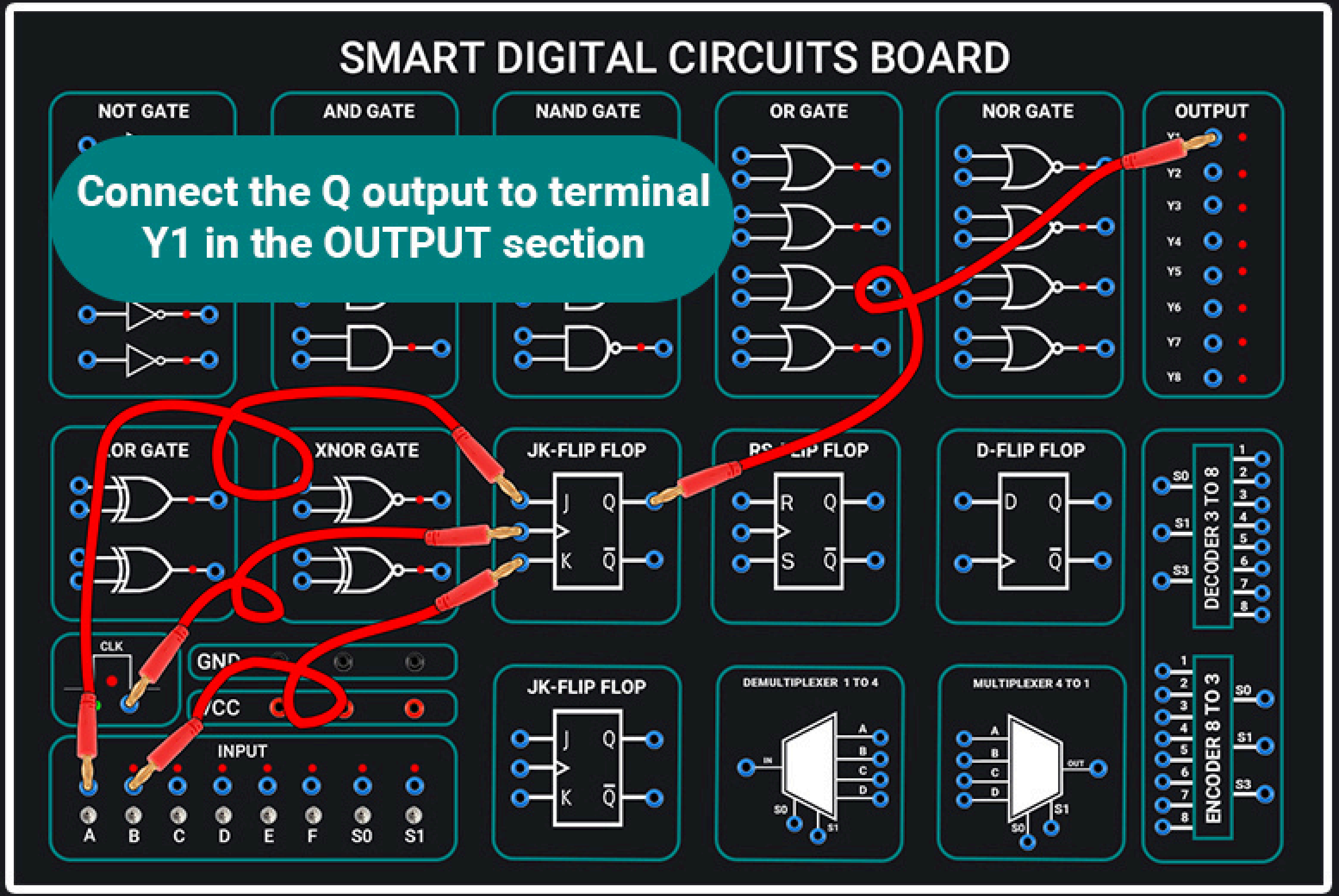
2


SMART DIGITAL CIRCUITS BOARD
Connect this terminal to the CLK pulse generator at the bottom left of the board

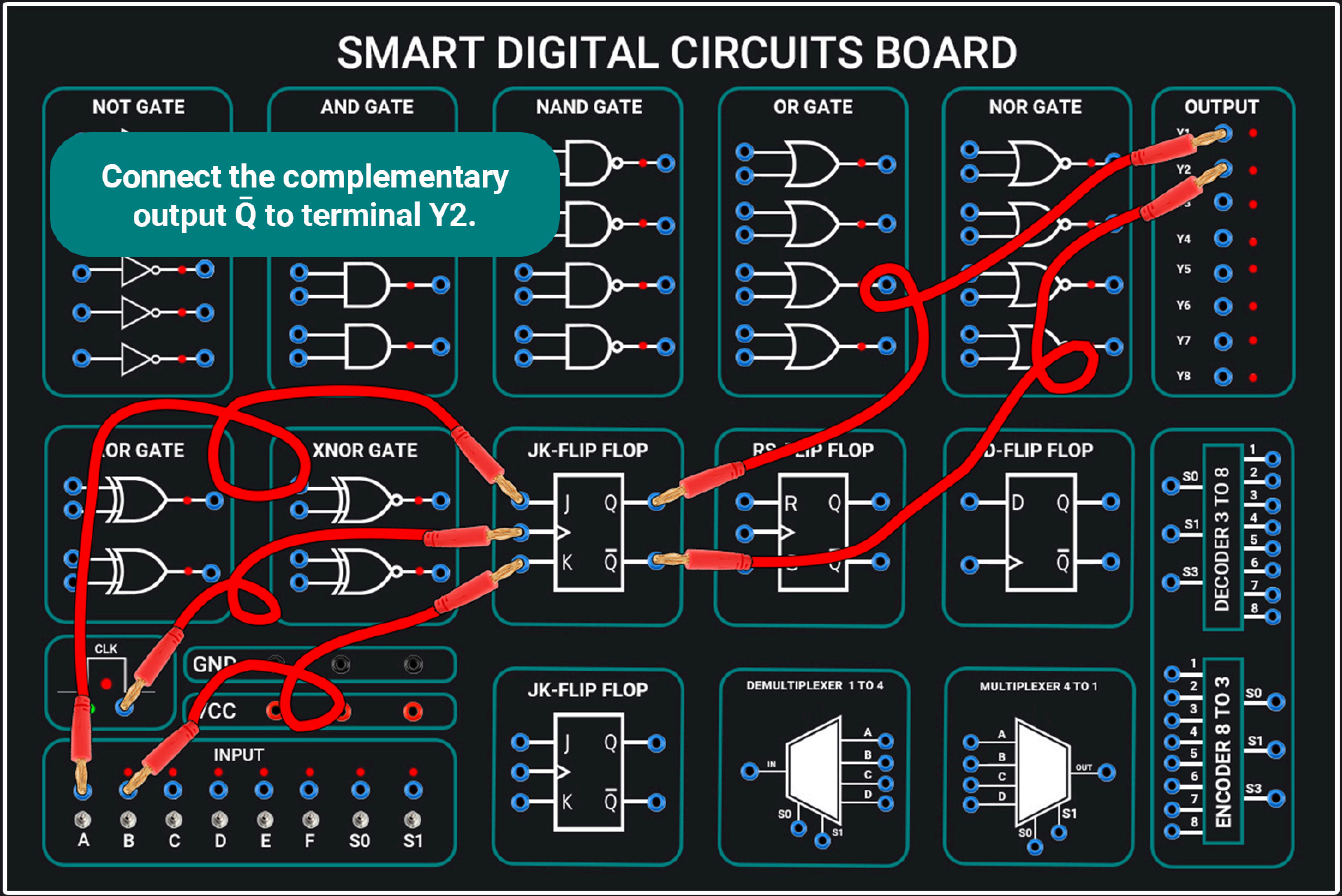
3


SMART DIGITAL CIRCUITS BOARD
Connect the K input terminal (bottom left blue pin) to switch B in the INPUT section.

4


SMART DIGITAL CIRCUITS BOARD
Connect the Q output to terminal Y1 in the OUTPUT section

5


SMART DIGITAL CIRCUITS BOARD
Connect the complementary output Q̄ to terminal Y2.
NOT GATE
AND GATE
NAND GATE
OR GATE
NOR GATE
OUTPUT
Y2
Y4
Y5
Y6
Y7
Y8
XNOR GATE
JK-FLIP FLOP
J
Q
K
Q̄
R
Q
D-FLIP FLOP
D
Q
Q̄
S0
S1
S3
DECODER 3 TO 8
1
2
3
4
5
6
7
8
CLK
GND
INPUT
A
B
C
D
E
F
S0
S1
JK-FLIP FLOP
J
Q
K
Q̄
DEMULTIPLEXER 1 TO 4
IN
A
B
C
D
S0
S1
MULTIPLEXER 4 TO 1
A
B
C
D
OUT
S0
S1
1
2
3
4
5
6
7
8
ENCODER 8 TO 3
S0
S1
S3